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EXPERIMENT 4

Aim: To perform morphological analysis.

Theory:

Morphological analysis in Natural Language Processing (NLP) involves examining the structure of words to identify their roots, prefixes, suffixes, and other morphological components. Using libraries like **spaCy** and **NLTK**, this process can be efficiently carried out. **spaCy** provides built-in support for tokenization, lemmatization, and part-of-speech tagging, helping to reduce words to their base forms (lemmas) and identify their grammatical features. **NLTK** also offers tools for morphological analysis, such as stemming and lemmatization, but with a focus on more traditional methods like the **Porter Stemmer**. Both libraries allow for efficient analysis of word forms, enabling tasks like text normalization and syntactic parsing, which are crucial for downstream NLP tasks such as text classification or named entity recognition.

Steps to perform morphological analysis

1. **Install spaCy:**
First, you need to install spaCy and download the relevant language model (e.g., en_core_web_sm for English).
2. **Load the Language Model:**
Load the spaCy language model to process the text.
3. **Tokenization:**
Tokenize the input text to break it into words and punctuation marks.
4. **Lemmatization:**
Use spaCy's lemmatizer to reduce each word to its base form (lemma).
5. **Part-of-Speech Tagging:**
Extract the part-of-speech (POS) for each token, which helps in understanding its grammatical role.
6. **Morphological Features:**
Access additional morphological features like tense, number, or gender, using token attributes.

Using NLTK:

1. **Install NLTK:**
Install NLTK and download necessary resources (e.g., stopwords, wordnet).
2. **Tokenization:**
Tokenize the input text into words using NLTK's tokenizer.
3. **Stemming:**
Use NLTK's stemming algorithms, such as the **Porter Stemmer**, to reduce words to their root form.



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4. **Lemmatization:**

NLTK provides **WordNetLemmatizer**, which reduces words to their base form using WordNet.

5. **Part-of-Speech Tagging:**

Tag words with their parts of speech to identify grammatical categories.

6. **Morphological Features (WordNet):**

NLTK's **WordNet** can be used to access definitions, synonyms, and antonyms, which is useful for understanding word meaning and usage.

Code:

1)

```
import spacy
nlp=spacy.load("en_core_web_sm")
```

```
text = "The quick brown fox jumps over the lazy dog."

doc = nlp(text)

for token in doc:
    print(f"Word: {token.text}")
    print(f"Lemma: {token.lemma_}")
    print(f"POS: {token.pos_}")
    print(f"Tag: {token.tag_}")
    print(f"Dependency: {token.dep_}")
    print(f"Is Stopword: {token.is_stop}")
    print("-" * 50)
```

2)

```
!pip install nltk
import nltk
nltk.download('wordnet')
```

```
from nltk.corpus import wordnet as wn

verb = "run"

synsets = wn.synsets(verb, pos=wn.VERB)

for syn in synsets:
    for lemma in syn.lemmas():
        print(f"Generated word: {lemma.name()}")
```



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Output

1)

```
Word: The  
Lemma: the  
POS: DET  
Tag: DT  
Dependency: det  
Is Stopword: True
```

```
-----  
Word: quick  
Lemma: quick  
POS: ADJ  
Tag: JJ  
Dependency: amod  
Is Stopword: False
```

```
-----  
Word: brown  
Lemma: brown  
POS: ADJ  
Tag: JJ  
Dependency: amod  
Is Stopword: False
```

```
-----  
Word: fox  
Lemma: fox  
POS: NOUN  
Tag: NN  
Dependency: nsubj  
Is Stopword: False
```

```
-----  
Word: jumps  
Lemma: jump  
POS: VERB
```



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```
Word: jumps
Lemma: jump
POS: VERB
Tag: VBZ
Dependency: ROOT
Is Stopword: False
```

```
-----
Word: over
Lemma: over
POS: ADP
Tag: IN
Dependency: prep
Is Stopword: True
```

```
-----
Word: the
Lemma: the
POS: DET
Tag: DT
Dependency: det
Is Stopword: True
```

```
-----
Word: lazy
Lemma: lazy
POS: ADJ
Tag: JJ
Dependency: amod
Is Stopword: False
```

```
-----
Word: dog
Lemma: dog
POS: NOUN
Tag: NN
Dependency: pobj
Is Stopword: False
```

```
-----
Word: .
Lemma: .
```

2)



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```
→ Generated word: run
Generated word: scat
Generated word: run
Generated word: scarper
Generated word: turn_tail
Generated word: lam
Generated word: run_away
Generated word: hightail_it
Generated word: bunk
Generated word: head_for_the_hills
Generated word: take_to_the_woods
Generated word: escape
Generated word: fly_the_coop
Generated word: break_away
Generated word: run
Generated word: go
Generated word: pass
Generated word: lead
Generated word: extend
Generated word: operate
Generated word: run
Generated word: run
Generated word: go
Generated word: run
Generated word: flow
Generated word: feed
Generated word: course
Generated word: function
Generated word: work
Generated word: operate
Generated word: go
Generated word: run
Generated word: range
Generated word: run
Generated word: campaign
Generated word: run
Generated word: play
Generated word: run
Generated word: run
```



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```
→ Generated word: prevail
Generated word: persist
Generated word: die_hard
Generated word: run
Generated word: endure
Generated word: run
Generated word: run
Generated word: execute
Generated word: carry
Generated word: run
Generated word: run
Generated word: guide
Generated word: run
Generated word: draw
Generated word: pass
Generated word: run
Generated word: lead
Generated word: run
Generated word: run
Generated word: black_market
Generated word: run
Generated word: run
Generated word: bleed
Generated word: run
Generated word: run
Generated word: run
Generated word: run_for
Generated word: run
Generated word: run
Generated word: consort
Generated word: run
Generated word: run
Generated word: ply
Generated word: run
Generated word: hunt
Generated word: run
Generated word: hunt_down
Generated word: track_down
Generated word: race
Generated word: run
Generated word: move
Generated word: go
Generated word: run
Generated word: melt
Generated word: run
```



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```
Generated word: run  
Generated word: melt_down  
Generated word: ladder  
Generated word: run  
Generated word: run  
Generated word: unravel
```

Conclusion:

Thus we successfully implemented morphological analysis using NLP.